BRAEDEN MARCHANT

Mechatronics Engineering

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EDUCATION

McMaster University | 2022-2027

Bachelor of Mechatronics Engineering (Co-Op)

SKILLS

- **CAD/PCB Design:** Autocad, Solidworks, Inventor, Fusion360, CATIA V5, 3D Printing, KiCad
- **Programming & Embedded Systems:** Python, C/C++, MATLAB/Simulink, Verilog/VHDL, STM32, ESP32, FPGA, Linux
- **Controls & Automation:** Mitsubishi PLC, Siemens PLC, Safety PLC, HMI Design, SCADA, PID Control
- **Electronics & Signal Processing:** Analog and Digital Circuit Design, Signal Processing, EMG
- **Machine Vision & AI:** Cognex, MVTec HALCON, Deep Learning, OpenCV, PyTorch
- **Project Management & Analysis:** Budget Tracking, Root Cause Analysis, Statistical Process Control (SPC)
- **Certifications:** Working at Heights, First Aid, Confined Space

WORK EXPERIENCE

Honda | May 2025 - August 2026

Electrical Engineering Intern (Paint 2 - CAST)

- Project lead on multiple $100K projects upgrading PLC systems, gaining hands-on experience in PLC programming, HMI design, wiring, electrical drawings, and project management.
- Project lead on Cognex vision system, used to detect underbody sealer presence, preventing multiple production downtime scenarios.
- Supported multiple AI vision systems using HALCON Deep Learning for automated model and jig detection.
- Supported multiple safety projects installing light curtains, area scanners, and safety limit switches. Gained safety PLC experience with Mitsubishi and Siemens.
- Installed a variety of sensors and electrical equipment such as: VFDs, RFID readers, pressure sensors, laser sensors, PID controllers.

Alectra Utilities | May - August 2024

Station Sustainment / Protection & Control

- Performed electrical diagnostic testing on equipment across multiple substations, supporting assessment and maintenance activities.
- Installed and wired substation equipment, including lightning arresters, circuit breakers, and monitoring devices.
- Assembled and wired transformer panels using various electrical components and hand tools.
- Collaborated with the P&C team to troubleshoot wide-area Ethernet networks and SCADA system issues.

Canada's Wonderland | June - August 2023

Mechanical Technician

- Performed daily preventative maintenance and safety inspections on roller coasters and amusement rides, identifying and correcting mechanical issues to ensure safe and reliable operation.
- Utilized a wide range of hand tools, power tools, & machine shop equipment to support maintenance and repair.
- Diagnosed, repaired, and rebuilt mechanical components including pneumatic cylinders, pistons, wheels, braking systems, and other ride-critical equipment, minimizing downtime and improving reliability.

PROJECT/TEAM EXPERIENCE

- **Cobot - Programming Lead | 2025-2026**
  Python, OpenCV, RTDE, Wiring, Robot Teach
- **MDL - Designathon | 2024/2025**
  CAD, Product Design
- **Krystal Klear Robotic Mirror Arm | 2023**
  Arduino, Wiring, CAD/ 3D Printing
- **Hexpad - Custom Macropad | 2026**
  PCB Design, Soldering, CAD / 3D Printing
- **Custom Micro FPV Drone | 2026 - Ongoing**
  Electronics, Soldering, RF, 3D Printing
- **FRC - Engineering Team Lead | 2019-2022**
  CAD, Wiring, Hand Tools